Experiment No 2: Program based on class and objects

Name of student : Shreya Dipak Patil

Roll No.2015 URN: 23031038 Class : SY A

Problem Statement to solve:

1. Write a program to print the area of a rectangle by creating a class named 'Area' having two methods. First method named as 'setDim' takes length and breadth of rectangle as parameters and the second method named as 'getArea' returns the area of the rectangle. Length and breadth of rectangle are entered through keyboard.

```
import java.util.*;

public class Area{
	double length, breadth;

	void setDim(double length, double breadth){
		this.length = length;
		this.breadth = breadth;
	}

	double getArea(){
		double area = length*breadth;
		return area;
	}

	public static void main(String args[]){
		Scanner sc = new Scanner(System.in);
		System.out.print("Enter length : ");
		double length = sc.nextFloat();
		System.out.print("Enter breadth : ");
		double breadth = sc.nextFloat();

		Area rectangle = new Area();
		rectangle.setDim(length, breadth);
		double area = rectangle.getArea();
		System.out.println("Area of rectangle is : "+area);
	}
}
```

```
C:\Java programs>javac Area.java

C:\Java programs>java Area
Enter length : 5
Enter breadth : 4
Area of rectangle is : 20.0

C:\Java programs>
```

2.Create a class named 'Student' with String variable 'name' and integer variable 'roll_no'. Assign the value of roll_no as '2' and that of name as "John" by creating an object of the class Student.

```java
public class Student{
        String name;
        int roll_no;

        public static void main(String args[]){
                Student s1 = new Student();
                s1.name = "John";
                s1.roll_no = 2;

                System.out.println("Student name : "+s1.name);
                System.out.println("Student roll_no : "+s1.roll_no);
        }
}
```

```
C:\Java programs>javac Student.java

C:\Java programs>java Student
Student name : John
Student roll_no : 2

C:\Java programs>
```

3.Assign and print the roll number, phone number and address of two students having names "Sam" and "John" respectively by creating two objects of class 'Student'.

```java
public class Students{
        String name;
        int roll_no;
        String phone_no;
        String address;

        public static void main(String args[]){
                Students s1 = new Students();
                s1.name = "John";
                s1.roll_no = 1;
                s1.phone_no = "1234567890";
                s1.address = "123 xyz street, Sangli";

                Students s2 = new Students();
                s2.name = "Sam";
                s2.roll_no = 2;
                s2.phone_no = "9876543210";
                s2.address = "123 abc street, Kolhapur";

                System.out.println("Student s1 name : "+s1.name);
                System.out.println("Student s1 roll_no : "+s1.roll_no);
                System.out.println("Student s1 Phone_no : "+s1.phone_no);
                System.out.println("Student s1 address : "+s1.address+"\n");

                System.out.println("Student s2 name : "+s2.name);
                System.out.println("Student s2 roll_no : "+s2.roll_no);
                System.out.println("Student s2 Phone_no : "+s2.phone_no);
                System.out.println("Student s2 address : "+s2.address);
        }
}
```

```
C:\Java programs>javac Students.java

C:\Java programs>java Students
Student s1 name : John
Student s1 roll_no : 1
Student s1 Phone_no : 1234567890
Student s1 address : 123 xyz street, Sangli

Student s2 name : Sam
Student s2 roll_no : 2
Student s2 Phone_no : 9876543210
Student s2 address : 123 abc street, Kolhapur

C:\Java programs>
```

4.Write a program to print the area and perimeter of a triangle having sides of 3, 4 and 5 units by creating a class named 'Triangle' without any parameter in its constructor.

```
public class Triangle{
	int a,b,c;

	Triangle(){
		a = 3;
		b = 4;
		c = 5;
	}

	int getPerimeter(){
		return a+b+c;
	}

	double getArea(){
		double s = (a+b+c)/2.0;
		return Math.sqrt(s * (s - a) * (s - b) * (s - c));
	}

	public static void main(String args[]){
		Triangle t1 = new Triangle();

		System.out.println("Perimeter of triangle : "+t1.getPerimeter());
		System.out.println("Area of triangle : "+t1.getArea());
	}
```

```
C:\Java programs>javac Triangle.java

C:\Java programs>java Triangle
Perimeter of triangle : 12
Area of triangle : 6.0

C:\Java programs>
```

5.Write a program to print the area and perimeter of a triangle having sides of 3, 4 and 5 units by creating a class named 'Triangle' with constructor having the three sides as its parameters.

```java
public class Triangle2{
	int a,b,c;
	Triangle2(int a, int b, int c){
		this.a = a;
		this.b = b;
		this.c = c;
	}

	int getPerimeter(){
		return a+b+c;
	}

	double getArea(){
		double s = (a+b+c)/2.0;
		return Math.sqrt(s * (s - a) * (s - b) * (s - c));
	}

	public static void main(String args[]){
		Triangle2 t1 = new Triangle2(3,4,5);

		System.out.println("Perimeter of triangle : "+t1.getPerimeter());
		System.out.println("Area of triangle : "+t1.getArea());
	}
}
```

```
C:\Java programs>javac Triangle2.java

C:\Java programs>java Triangle2
Perimeter of triangle : 12
Area of triangle : 6.0

C:\Java programs>
```

6.Write a program to print the area of two rectangles having sides (4,5) and (5,8) respectively by creating a class named 'Rectangle' with a method named 'Area' which returns the area and length and breadth passed as parameters to its constructor.

```java
public class Rectangle {
    int length, breadth;

    Rectangle(int l, int b) {
        length = l;
        breadth = b;
    }

    int Area() {
        return length * breadth;
    }

    public static void main(String args[]) {

        Rectangle rect1 = new Rectangle(4, 5);
        Rectangle rect2 = new Rectangle(5, 8);

        System.out.println("Area of first rectangle: " + rect1.Area());
        System.out.println("Area of second rectangle: " + rect2.Area());
    }
}
```

```
C:\Java programs>javac Rectangle.java

C:\Java programs>java Rectangle
Area of first rectangle: 20
Area of second rectangle: 40

C:\Java programs>
```

7.Write a program to print the area of a rectangle by creating a class named 'Area' taking the values of its length and breadth as parameters of its constructor and having a method named 'returnArea' which returns the area of the rectangle. Length and breadth of rectangle are entered through keyboard.

```java
import java.util.*;

public class AreaR{
    int length, breadth;

    AreaR(int l, int b) {
        length = l;
        breadth = b;
    }

    int Area() {
        return length * breadth;
    }

    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter length of first rectangle : ");
        int l1 = sc.nextInt();
        System.out.print("Enter breadth of first rectangle : ");
        int b1 = sc.nextInt();
```

```
        System.out.print("Enter length of second rectangle : ");
        int l2 = sc.nextInt();
        System.out.print("Enter breadth of second rectangle : ");
        int b2 = sc.nextInt();

        AreaR rect1 = new AreaR(l1, b1);
        AreaR rect2 = new AreaR(l2, b2);

        System.out.println("Area of first rectangle: " + rect1.Area());
        System.out.println("Area of second rectangle: " + rect2.Area());
    }
}
```

```
C:\Java programs>javac AreaR.java

C:\Java programs>java AreaR
Enter length of first rectangle : 4
Enter breadth of first rectangle : 5
Enter length of second rectangle : 6
Enter breadth of second rectangle : 7
Area of first rectangle: 20
Area of second rectangle: 42

C:\Java programs>
```

8.Print the average of three numbers entered by user by creating a class named 'Average' having a method to calculate and print the average.

```
import java.util.*;

public class Average {

    void calculateAverage(int num1, int num2, int num3) {
        double average = (num1 + num2 + num3) / 3.0;
        System.out.println("The average of the three numbers is: " + average);
    }

    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the first number: ");
        int num1 = sc.nextInt();
        System.out.print("Enter the second number: ");
        int num2 = sc.nextInt();
        System.out.print("Enter the third number: ");
        int num3 = sc.nextInt();

        Average avg = new Average();

        avg.calculateAverage(num1, num2, num3);
    }
}
```

9.Print the sum, difference and product of two complex numbers by creating a class named 'Complex' with separate methods for each operation whose real and imaginary parts are entered by user.

```java
import java.util.*;

public class Complex {
    int real, imaginary;

    Complex() {
        real = 0;
        imaginary = 0;
    }

    void setComplex(int real, int imaginary) {
        this.real = real;
        this.imaginary = imaginary;
    }

    static Complex add(Complex c1, Complex c2) {
        Complex result = new Complex();
        result.real = c1.real + c2.real;
        result.imaginary = c1.imaginary + c2.imaginary;
        return result;
    }

    static Complex subtract(Complex c1, Complex c2) {
        Complex result = new Complex();
        result.real = c1.real - c2.real;
        result.imaginary = c1.imaginary - c2.imaginary;
        return result;
    }

    static Complex multiply(Complex c1, Complex c2) {
        Complex result = new Complex();
        result.real = c1.real * c2.real - c1.imaginary * c2.imaginary;
        result.imaginary = c1.real * c2.imaginary + c1.imaginary * c2.real;
        return result;
    }

    void print() {
        System.out.println(real + " + " + imaginary + "i");
    }

    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        Complex complex1 = new Complex();
        Complex complex2 = new Complex();

        System.out.print("Enter the real part of the first complex number: ");
        complex1.real = sc.nextInt();
        System.out.print("Enter the imaginary part of the first complex number: ");
        complex1.imaginary = sc.nextInt();

        System.out.print("Enter the real part of the second complex number: ");
        complex2.real = sc.nextInt();
        System.out.print("Enter the imaginary part of the second complex number: ");
        complex2.imaginary = sc.nextInt();

        Complex sum = add(complex1, complex2);
        Complex difference = subtract(complex1, complex2);
        Complex product = multiply(complex1, complex2);

        System.out.print("Sum: ");
        sum.print();
        System.out.print("Difference: ");
        difference.print();
        System.out.print("Product: ");
        product.print();
    }
}
```

```
C:\Java programs>javac Complex.java

C:\Java programs>java Complex
Enter the real part of the first complex number: 9
Enter the imaginary part of the first complex number: 8
Enter the real part of the second complex number: 1
Enter the imaginary part of the second complex number: 2
Sum: 10 + 10i
Difference: 8 + 6i
Product: -7 + 26i

C:\Java programs>
```

10. Write a program that would print the information (name, year of joining, salary, address) of three employees by creating a class named 'Employee'. The output should be as follows:

Name	Year of joining	Address
Robert	1994	64C- WallsStreat
Sam	2000	68D- WallsStreat
John	1999	26B- WallsStreat

```java
public class Employee {
    String name;
    int yearOfJoining;
    double salary;
    String address;

    Employee(String name, int yearOfJoining, double salary, String address) {
        this.name = name;
        this.yearOfJoining = yearOfJoining;
        this.salary = salary;
        this.address = address;
    }

    void printDetails() {
        System.out.println(name + "\t\t" + yearOfJoining + "\t\t" + salary + "\t\t" + address);
    }

    public static void main(String[] args) {
        Employee emp1 = new Employee("Robert", 1994, 50000, "64C- WallsStreat");
        Employee emp2 = new Employee("Sam", 2000, 60000, "68D- WallsStreat");
        Employee emp3 = new Employee("John", 1999, 55000, "26B- WallsStreat");

        System.out.println("Name\t\tYear of joining\tSalary\t\tAddress");
        emp1.printDetails();
        emp2.printDetails();
        emp3.printDetails();
    }
}
```

```
C:\Java programs>javac Employee.java

C:\Java programs>java Employee
Name            Year of joining Salary          Address
Robert          1994            50000.0         64C- WallsStreat
Sam             2000            60000.0         68D- WallsStreat
John            1999            55000.0         26B- WallsStreat

C:\Java programs>
```